

BUSINESS CASE STUDY

Brazil E-Commerce Performance

Turning Marketplace Transaction Data into Strategic Growth Recommendations

Dataset: Olist Brazilian E-Commerce Public Dataset

Tools: PostgreSQL • SQL • Tableau

Prepared as a Business Analyst / BI Portfolio Project

01 Executive Summary

This case study analyzes marketplace performance for a Brazilian e-commerce platform using the Olist public transaction dataset. A relational database was built in PostgreSQL, analytical SQL queries were developed to measure category, seller, customer, and logistics performance, and the results were consolidated into an executive Tableau dashboard. The objective was to identify where growth is concentrated, where operational risk exists, and what actions the business should take next.

\$13.22M Total GMV	96.48K Total Orders	93.36K Customers	2,970 Sellers	\$137.04 Avg Order Value	\$19.95 Avg Shipping Fee
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Three findings anchor the recommendations in this report:

- **Revenue is concentrated** in a small set of categories, sellers, and one dominant state (São Paulo), which drives scale but also creates dependency risk.
- **Delivery speed is a measurable driver of customer satisfaction** — longer delivery windows correlate with lower review scores.
- **Higher-value states are not the highest-volume states**, revealing an unexploited opportunity to grow average order value outside the core São Paulo market.

02 Business Background

Olist operates a multi-seller marketplace that connects small and mid-sized Brazilian retailers with customers across the country, similar in structure to Amazon Marketplace or Mercado Livre. The platform earns revenue as sellers list products and fulfil orders through Olist's logistics network. Because Olist depends on thousands of independent sellers rather than owning inventory itself, marketplace health depends on three things staying aligned: product demand across categories, seller reliability, and delivery performance across a geographically large and logistically uneven country.

This analysis takes the perspective of a Business Analyst supporting the marketplace operations and strategy team, using historical order data to evaluate current performance and surface where management attention should be focused.

03 Business Questions

The analysis was scoped around five questions the operations and strategy team needed answered:

- Which product categories generate the most revenue, and which generate the most orders? Are they the same categories?
- How concentrated is seller performance — do a few sellers account for most of the GMV?
- Does delivery speed influence customer review scores, and by how much?
- Which states drive the most total revenue, and which states have the highest-value customers?
- Where should the business focus resources to grow GMV while protecting customer satisfaction?

04 Data & Methodology

The underlying dataset is the Olist Brazilian E-Commerce Public Dataset, containing anonymized records of approximately 96,500 orders placed between 2016 and 2018, including order items, products, sellers, customers, reviews, payments, and geolocation data.

Workflow

Raw CSV extracts were loaded into a normalized **PostgreSQL** database and joined into a single analytical model spanning orders, order items, products, sellers, customers, and reviews. Six SQL scripts were developed to answer the business questions above, covering executive KPIs, category performance, seller concentration (Pareto analysis), customer geography, and delivery/logistics performance. Query outputs were connected to **Tableau** to build an interactive executive dashboard.

Analysis Area	What It Measures	SQL Script
Executive KPIs	GMV, orders, customers, sellers, AOV, average shipping fee	01_executive_kpi.sql
Category Analysis	GMV, order volume, and average price by product category	02_category_analysis.sql
Seller Analysis	Seller-level GMV vs. average review score	03_seller_analysis.sql
Customer Analysis	GMV and average order value by state	04_customer_analysis.sql
Logistics Analysis	Average delivery days vs. average review score	05_logistics_analysis.sql
Pareto Analysis	Cumulative GMV share ranked by seller	06_pareto_analysis.sql

Key metric definitions: GMV (Gross Merchandise Value) is the total value of items sold. AOV (Average Order Value) is GMV divided by total orders. Average Shipping Fee is freight revenue divided by total orders. Review Score is the customer's 1–5 satisfaction rating per order.

05 Dashboard Overview

All KPIs and analyses were consolidated into a single-page executive Tableau dashboard, giving stakeholders category, seller, logistics, and regional performance in one view.

Brazil E-commerce Performance Dashboard

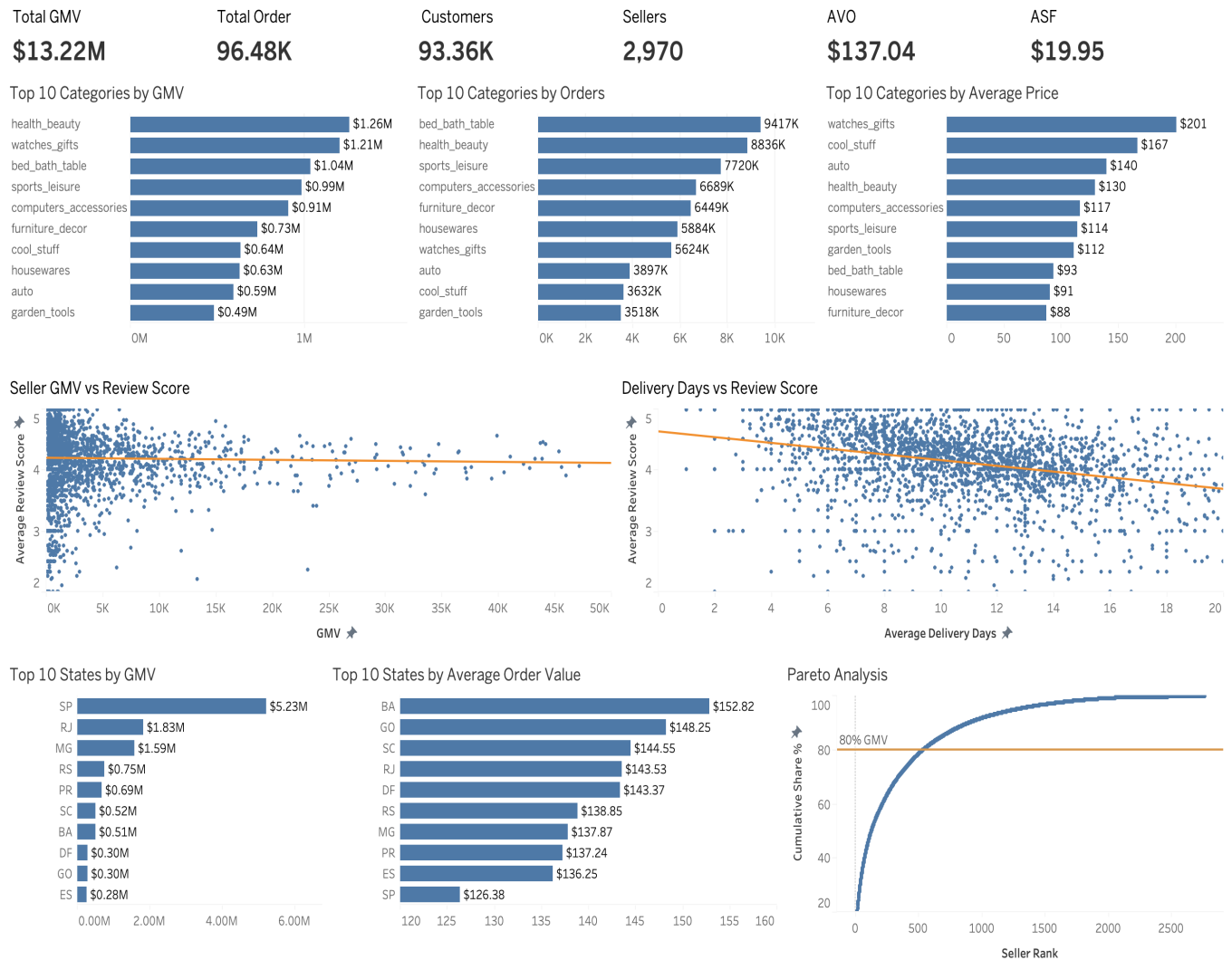


Figure 1. Brazil E-Commerce Performance Dashboard (Tableau)

06 Key Insights

6.1 Revenue Leaders Are Not Always Volume Leaders

Health & Beauty leads in GMV (\$1.26M) but ranks second in order volume, while Bed, Bath & Table leads in order count (9,417 orders) but ranks third in GMV. Watches & Gifts stands out as a premium category: it ranks second in GMV despite mid-table order volume, because it carries the highest average price in the marketplace at \$201 per item, well above the next-highest category (Cool Stuff at \$167). This signals two distinct growth levers: driving basket volume in high-frequency categories such as Bed, Bath & Table, and driving margin through premium categories such as Watches & Gifts.

6.2 Seller Revenue Is Highly Concentrated

The Pareto analysis shows that roughly the top 500 sellers — out of 2,970 total, about 17% of the seller base — already account for close to 80% of total GMV, and the cumulative curve flattens sharply after that point. This concentration means marketplace revenue is structurally dependent on a relatively small group of top-performing sellers, which is efficient for platform management but represents a concentration risk if any of those sellers underperform or churn.

6.3 Delivery Speed Is a Measurable Driver of Satisfaction

There is a clear negative relationship between average delivery days and average review score: orders delivered in the fastest windows cluster near 4.7–5.0 stars, while orders taking 14 days or longer trend down toward 3.7–4.0 stars. By contrast, seller GMV shows almost no relationship with review score — high- and low-revenue sellers both span the full satisfaction range. This indicates that logistics performance, not seller size, is the more actionable lever for protecting customer satisfaction.

6.4 Regional Value Is Split Between Volume and Price

São Paulo (SP) is the dominant market by total GMV (\$5.23M, roughly 3x the next largest state, Rio de Janeiro at \$1.83M), reflecting its population and logistics density. However, SP has the lowest average order value among the top ten states at \$126.38, while smaller markets such as Bahia (\$152.82), Goiás (\$148.25), and Santa Catarina (\$144.55) show meaningfully higher average spend per order. This suggests São Paulo drives the business through volume, while several mid-sized states represent an under-scaled but higher-value customer base.

07 Business Recommendations

1. Protect and formalize the top-seller tier

Given that roughly 500 sellers drive ~80% of GMV, introduce a tiered seller-success program (dedicated support, performance monitoring, early churn warning) for this group to protect the revenue base, while running a separate onboarding/growth track for long-tail sellers.

2. Set delivery-time SLAs tied to review-score targets

Because review scores decline consistently once delivery exceeds ~10–12 days, set a target delivery ceiling (e.g., under 10 days) for priority categories and regions, and prioritize logistics investment in states where delivery times currently exceed that threshold.

3. Grow basket volume in high-frequency categories

Invest marketing and cross-sell placement behind Bed, Bath & Table, Health & Beauty, and Sports & Leisure, which already combine strong order volume with strong GMV, to compound existing demand rather than build new categories from zero.

4. Expand share in high-AOV states outside São Paulo

Run targeted seller-recruitment and demand-generation in Bahia, Goiás, and Santa Catarina, where average order value is 15–20% higher than in São Paulo, to lift blended AOV as these regions scale rather than relying solely on São Paulo volume.

5. Use Watches & Gifts as a template for premium-category expansion

With an average price of \$201, more than double the marketplace average, Watches & Gifts shows that premium categories can materially lift GMV without proportional order growth. Evaluate adjacent premium categories (e.g., Cool Stuff, Auto) for similar merchandising and pricing treatment.